Karnaugh Maps

Lecture 6



Things we learned so far....

01 Number System

02 Logic Gates

03 Boolean Algebra

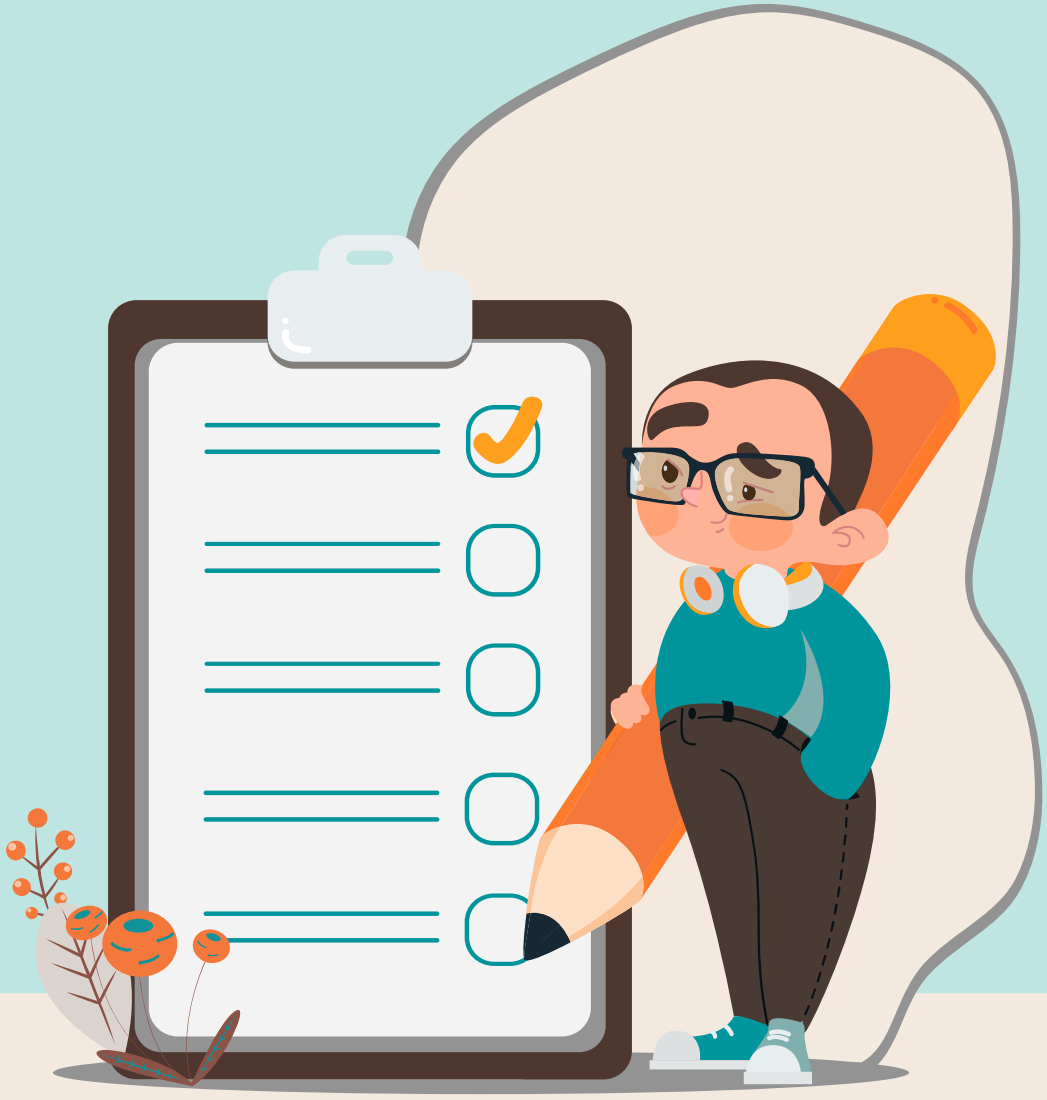
04 Boolean Functions

ICT is Over!!



01

Logic Minimization



Why Minimization?

- Providing a digital logic-based solution for a given practical problem, is a step-by-step process.
- First you need to understand the problem, define Boolean variables and obtain a truth table for the given problem.
- Once the truth table is available then you can write down logic expressions. (either in SOP form or POS form).

Why Minimization?

- If you straight away implement this logic expression without simplifying it, you may require a larger number of logic gates, therefore larger number of ICs in your circuit.
- Therefore, minimization of the logic expression is an important step in the process of logic design.

Logic Minimization Techniques

01

Boolean Algebra Laws

Simplification using Rules

02

Karnaugh Maps

Graphical Method

02

Karnaugh Map



Karnaugh Map

- This was invented by Maurice Karnaugh in 1953. (Also pronounced as K-Map)
- We can draw K-maps for 2 variables, 3 variables and 4 variables.
- If your logic expression has more than 4 variables (5,6,..) then there are other minimization techniques that is easier than K-maps.

Steps - Minimization using Karnaugh map

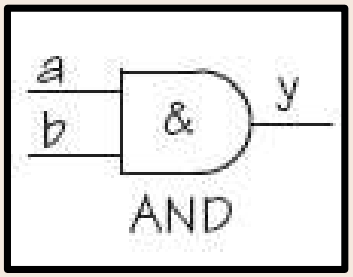
1. Transfer logic expression/ truth table into a K-Map.
2. Group 1’s in the power of 2.
3. Convert groupings into logic expression.
4. Minimization done!



2 variable K-map

- Two variables a and b → 2^2 = Four possibilities
- Consider the logic expression y=a.b

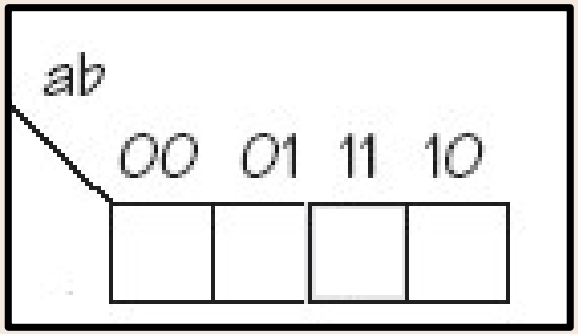
Logic Circuit



Truth Table

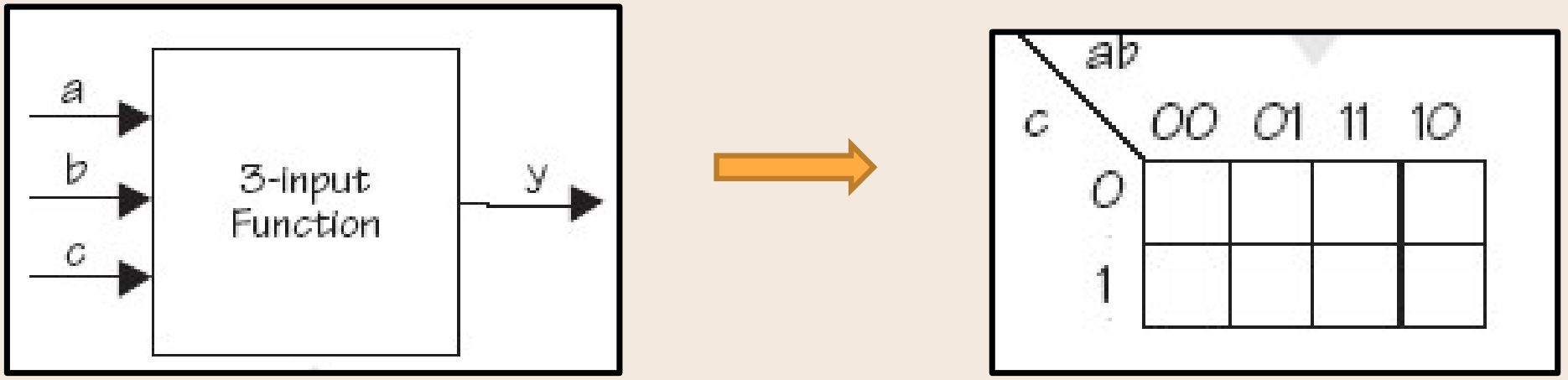
a	b	y
0	0	0
0	1	0
1	0	0
1	1	1

K-Map



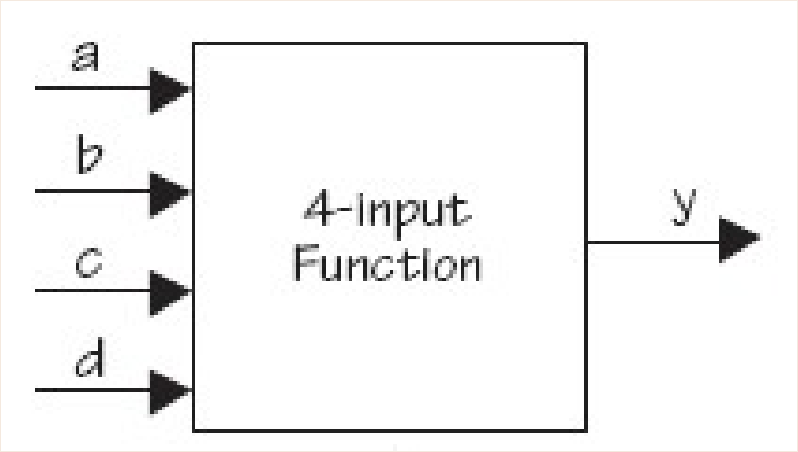
3 variable K-map

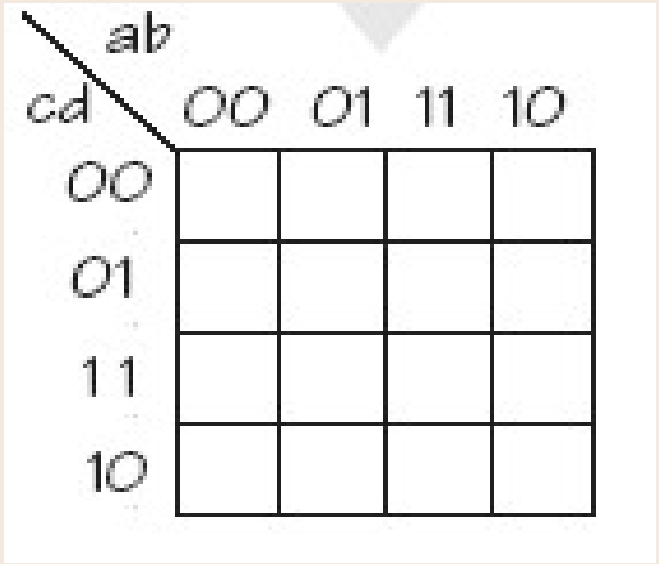
- When it is three variables a, b and c there are 8 possibilities (2^3).
- Therefore, the K-map will have 8 squares.



4 variable K-map

- 4 variables → 16 possibilities → 16 squares





03

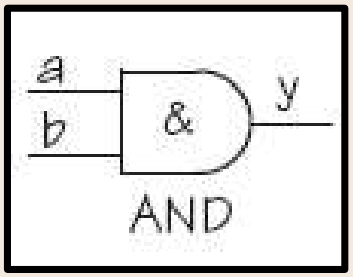
Filling K-Maps



2 variable K-map

- Two variables a and b → Four possibilities
- Consider the logic expression y=a.b

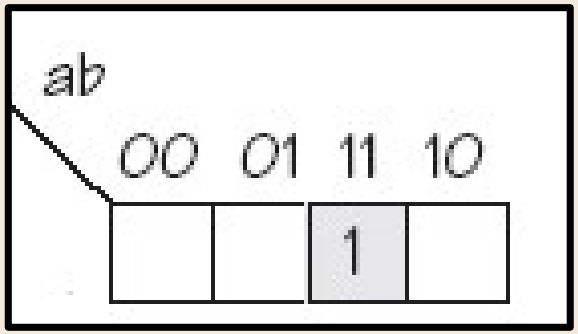
Logic Circuit



Truth Table

a	b	y
0	0	0
0	1	0
1	0	0
1	1	1

K-Map



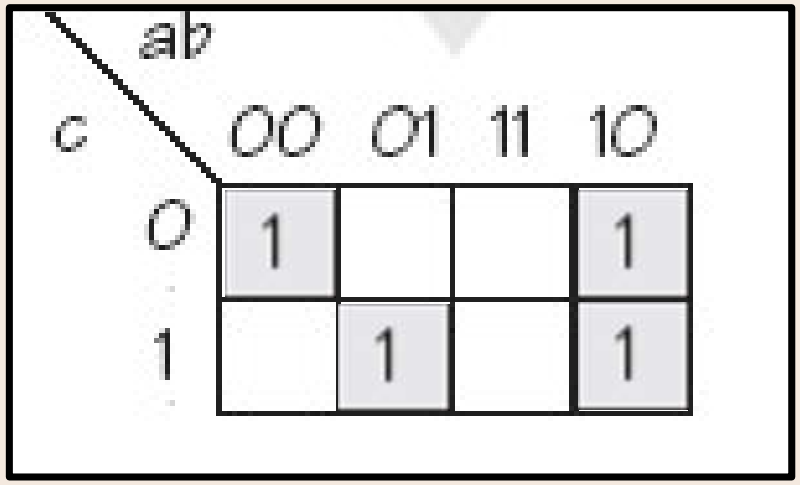
Example: Transfer this truth table to a suitable K-Map

a	b	c	y
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0



Example: Transfer this truth table to a suitable K-Map

a	b	c	y
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0



Question: Transfer the following logic function to a K- Map

$$y = a.b.c + a.b.\overline{c}$$



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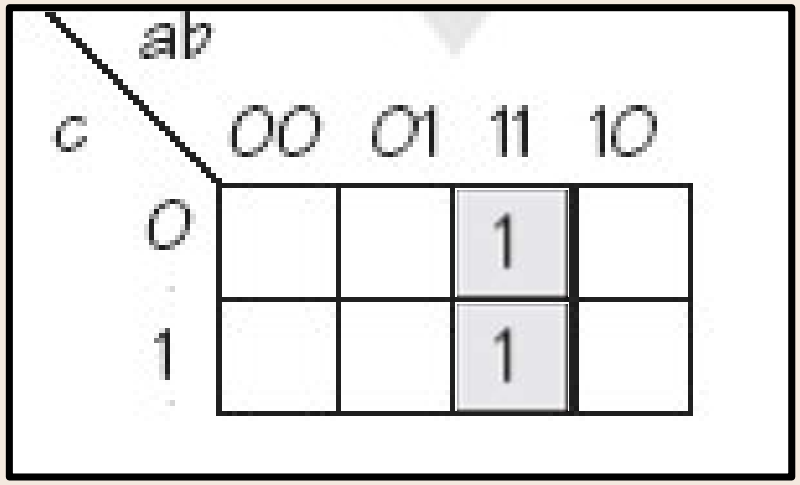


2 Methods

1. Truth Table -> K-Maps
2. SOP -> K-Maps

Method 1 : Transfer SOP to truth table and then to a K- Map

a	b	c	y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1



Method 2 : Direct Transfer from SOP

$$y = a.b.c + a.b.\overline{c}$$

